

Felix Hering

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Education

Georgia Institute of Technology, Atlanta, GA

Fall 2022 - Spring 2026

- B.S. in Mechanical Engineering, Minor in Robotics

GPA: 3.91/4.00

Work Experience

Electromechanical Systems Intern

Emory Healthcare, Atlanta, GA

Summer 2025 - Fall 2025

- Owned mechanical architecture and precision alignment of a multi-camera optomechanical system for radiation therapy guidance, maintaining <0.5 mm positional error across assembly, calibration, and operation.
- Implemented custom Python algorithms for stereo calibration, depth fusion, and real-time point cloud generation.
- Applied GD&T principles to constrain camera mounts and ensure repeatable alignment across assemblies

Mechanical Test & Validation Engineer (Co-op)

W.L. Gore & Associates Microwave Cable & Capacitors, Newark, DE

January 2024 - May 2024

- Designed test fixtures and environmental chambers for accelerated sulfur/humidity degradation of microwave cables; managed multi-factor test matrices and executed long-duration reliability studies.
- Performed mechanical and materials-level testing (pull tests, solderability assessment, surface chemistry analysis via XPS) and interpreted failure modes to drive new handling/storage protocols.
- Developed a Python data-analysis suite for RF/VSWR measurements from VNAs, improving test throughput and enabling standardized reporting across the floor.

Mechanical Systems Engineer (Research)

Water-Energy Research Lab, Atlanta, GA

Summer 2024

- Modeled and tested a coupled thermal-fluid mechanical system to optimize geometry, gap spacing, and thermal gradients for stable long-term efficiency
- Designed and executed experimental validation using LabVIEW DAQ, correlating sensor data with model predictions to inform design iteration
- Performed surface characterization of polymer membranes (AFM roughness, contact angle) to assess fouling behavior and performance degradation under operating conditions

Projects

Smart Vein Finder Imaging System

Fall 2025

- Designed multi-wavelength NIR imaging device housing with tight tolerances for aligned illumination and camera placement; fabricated using laser cutting and 3D printing
- Developed custom PCB for LED array driver circuits and synchronized camera control; implemented depth estimation algorithms on NVIDIA Jetson Nano
- Conducted optical and mechanical validation testing across varying skin tones and distances; iterated design based on alignment stability and assembly constraints

Gesture-Mimicking Robotic Arm

Fall 2025

- Built a 6-DOF serial robotic arm that replicates human arm motion using OpenCV for vision-based joint tracking.
- Developed kinematic mapping and servo interpolation algorithms to convert 2D keypoints into 3D joint angles
- Implemented embedded motor control in C on an STM32 microcontroller with a custom PCB

Simulations Lead at GT Supersonics Club

Fall 2024 - Spring 2025

- Designed Solidworks mechanical mounts for high-pressure/temperature instrumentation on propulsion test stand; performed shear and deflection hand calculations to prevent structural failure during dynamic tests
- Supported test stand development, sensor integration, and calibration for nozzle exit flow characterization
- Contributed to NPSS simulations and Python-based modeling to validate experimental thrust and flow data

Technical Skills

Hardware: CAD (SolidWorks, NX, Fusion 360), GD&T, PCB Design (KiCAD), Sensors & Instrumentation

Testing & Validation: LabVIEW, DAQ systems, Environmental & Mechanical Testing, Test Fixtures

Analysis & Simulation: Machine Design, Altair Inspire, NPSS, Simulink, Signal Processing

Embedded & Programming: Python, MATLAB, C, STM32, Raspberry Pi, Linux, Git, OpenCV, DepthAI